

PART-A (2 MARKS)

1. Define Data Science and write the benefits of Data Science.

Data Science is an interdisciplinary field that uses techniques from statistics, programming, and domain knowledge to collect, analyze, and interpret data in order to extract meaningful insights and support decision-making.

Benefits of Data Science:

- Helps in better decision-making through data-driven insights
- Enables prediction and forecasting (e.g., sales, weather)
- Improves efficiency and productivity in organizations
- Supports automation and intelligent systems
- Enhances customer experience through personalized recommendations
- Assists in identifying trends and patterns in large datasets

2. What is the role of data science in business, medical research, healthcare, education, social media, technology and financial institutions?

1. Business:

- Helps in decision-making using data insights
- Improves customer targeting and marketing strategies
- Enables sales forecasting and demand prediction

2. Medical Research:

- Assists in drug discovery and development
- Analyzes clinical trial data for better outcomes
- Identifies patterns in diseases for early detection

3. Healthcare:

- Supports diagnosis and treatment planning
- Predicts patient risks and improves patient care
- Helps in managing hospital resources efficiently

4. Education:

- Enables personalized learning for students
- Tracks student performance and improves outcomes
- Helps institutions in planning and decision-making

5. Social Media:

- Powers recommendation systems (posts, ads, friends)
- Analyzes user behavior and trends
- Detects fake news and harmful content

6. Technology:

- Drives artificial intelligence and machine learning applications
- Improves system performance and automation
- Enhances cybersecurity through threat detection

7. Financial Institutions:

- Detects fraud and suspicious transactions
- Assists in risk management and credit scoring
- Enables algorithmic trading and investment decisions

3. Write the main types/categories of data?

Structured Data

- Organized in a fixed format (rows and columns)
- Easy to store and analyze
- *Example:* Databases, spreadsheets

Unstructured Data

- No predefined format or structure
- Difficult to process directly
- *Example:* Images, videos, social media posts

Semi-Structured Data

- Does not follow a strict structure but has some organization
- Uses tags or markers
- *Example:* JSON, XML files

Qualitative Data (Categorical Data)

- Descriptive and non-numerical
- Represents categories or labels
- *Example:* Gender, color, names

Quantitative Data (Numerical Data)

- Numerical and measurable
- Used for calculations and analysis
- *Example:* Age, height, income

4. Compare and contrast structured data and unstructured data.

Aspect	Structured Data	Unstructured Data
Definition	Data organized in a fixed format (rows & columns)	Data without a predefined structure
Format	Tabular (tables, spreadsheets, databases)	Text, images, videos, audio, etc.
Storage	Stored in relational databases (RDBMS)	Stored in data lakes, file systems
Processing	Easy to process and analyze	Difficult to process; needs advanced tools
Flexibility	Less flexible (fixed schema)	Highly flexible (no fixed schema)
Examples	Excel sheets, SQL databases	Emails, social media posts, multimedia files

5. Write the applications of Data Science.

Healthcare: Disease prediction, medical image analysis, personalized treatment

Business & Marketing: Customer segmentation, sales forecasting, targeted advertising

Finance: Fraud detection, credit scoring, risk analysis, algorithmic trading

E-commerce: Recommendation systems, demand prediction, customer behavior analysis

Education: Personalized learning, student performance analysis

Social Media: Content recommendation, sentiment analysis, trend detection

Transportation: Traffic prediction, route optimization, ride-sharing systems

Manufacturing: Predictive maintenance, quality control, process optimization

Cybersecurity: Threat detection, anomaly detection, data protection

Entertainment: Movie/music recommendations, audience analysis

6. What is NLP? Is natural language structured data?

Natural Language Processing (NLP) is a branch of artificial intelligence that enables computers to understand, interpret, and generate human language (text or speech).

Example: Chatbots, language translation, sentiment analysis.

Is natural language structured data?

No, natural language is unstructured data because it does not follow a fixed format like tables or rows and columns. It is usually in the form of text, making it more complex to process.

7. What is machine generated data with an example?

Machine-generated data is data that is automatically created by devices, systems, or software without direct human input.

Example:

Sensor data from a smart device (like temperature readings from a weather sensor) or log files generated by a computer system.

8. What is graph-based or network data?

Graph-based (or network) data represents information as nodes (entities) and edges (relationships) between them.

Example: Social networks where users are nodes and friendships are edges.

9. List the steps involved in data science processing?

Data Collection

Data Cleaning (Preprocessing)

Exploratory Data Analysis (EDA)

Model Building

Model Evaluation

Deployment and Interpretation

10. Define outliers with examples.

Outliers are data points that significantly differ from the rest of the dataset.

Examples:

- In a class where most students score between 40–90, a score of 5 or 100 may be an outlier.
- In a dataset of ages (20–50), a value like 120 is an outlier.

In short: Outliers are unusual or extreme values that may affect analysis and results.

11. Define the process of removing duplicate records and handling missing values in a dataset.

The process is Data Cleaning (Preprocessing). It removes duplicates and handles missing values to improve data quality.

Example:

Dataset (Marks):

[85, 90, ?, 75, 80]

- Using mean method:
Mean = $(85 + 90 + 75 + 80) / 4 = 82.5$
Missing value → replaced with 82.5

Benefit:

- Improves data quality
- Ensures accurate analysis
- Removes errors and inconsistencies
- Enhances reliability of results

12. Define the term retrieving data.

Data Retrieval is the process of obtaining data from sources.

1. Internal Sources:

Data collected from within an organization.

Examples:

- Company databases (sales records)
- Employee records
- Customer transaction data

2. External Sources:

Data collected from outside the organization.

Examples:

- Websites and APIs
- Government reports
- Social media platforms

13. Define data exploration process.

Data Exploration is the process of analyzing datasets using statistics and visualization to understand patterns and relationships.

14. What are different facets (types) of data in Data Science?

Facets of data refer to different characteristics or dimensions of data.

Types:

- Structured

- Unstructured
- Natural language
- Machine-generated
- Graph-based
- Audio, video, and images
- Streaming

15. What is data preparation and its process or main tasks?

Data Preparation is converting raw data into usable form.

Main Tasks:

1. Data Cleaning
 - a. Removing duplicates
 - b. Handling missing values
 - c. Removing noise
2. Data Integration
 - a. Combining data from multiple sources
 - b. Resolving data conflicts
3. Data Transformation
 - a. Normalization
 - b. Aggregation
 - c. Feature selection

16. What is data modeling?

Data modeling is the process of creating a structure or model to represent data and its relationships for analysis and prediction.

Example:

A sales prediction model:

- Input data: Past sales, advertising cost, season
- Model: Regression model
- Output: Predicted future sales

17. Write about presentation and automation process.

Presentation: Communicating insights using charts/reports.

Automation: Using tools to perform tasks automatically (e.g., pipelines, scripts).

18. Write about the methods used to perform EDA.

1. Univariate analysis: Provides summary statistics for each field in the raw data set (or) summary only on one variable

2. Bivariate analysis is performed to find the relationship between each variable in the dataset and the target variable of interest (or) using two variables and finding a relationship between them.

3. Multivariate analysis is performed to understand interactions between different fields in the dataset (or) finding interactions between variables more than 2

19. Differentiate between Numerical Data and Categorical Data

Numerical Data:

Data in the form of numbers that can be measured and used for calculations.

Example: Age, salary, marks

Categorical Data:

Data that represents labels or categories, not numbers.

Example: Gender, color, department

20. Differentiate the nominal data and ordinal data with examples.

Nominal Data:

Categorical data with no specific order.

Example: Colors (red, blue, green), gender

Ordinal Data:

Categorical data with a meaningful order or ranking.

Example: Grades (A, B, C), satisfaction level (low, medium, high)

21 Give example of unstructured data science

Example of unstructured data: Images, videos, emails, social media posts

22 Mention the significance of setting goals in a data science project.

Setting goals helps in

clear direction,
better planning,
and accurate results.

23 What is a project charter? Show its Components.

Project Charter: Document defining project scope and objectives.

Components:

- Problem statement
- Objectives
- Scope
- Stakeholders
- Timeline

24 Write the fundamentals of EDA

Understanding the dataset (structure, variables)

Data cleaning (handling missing values, duplicates)

Descriptive statistics (mean, median, etc.)

Data visualization (charts, graphs)

Identifying patterns, trends, and outliers

25. Define the significance of EDA.

- Helps in understanding the structure and characteristics of data
- Identifies missing values, outliers, and errors
- Reveals patterns, trends, and relationships
- Supports better decision-making and model building

26. Define data ware housing, data mart and data lake

Data Warehouse: Central storage of structured data

Data Mart: Subset of data warehouse

Data Lake: Stores raw structured & unstructured data

27. Mention any 4 common errors that occur in data

Missing values

Duplicates

Incorrect data types

Outliers

28. What are the implications of erroneous data for analysis?

Erroneous data leads to

wrong analysis,
poor decisions,
and inaccurate models.

29. Write about the visual aids for EDA.

Visual aids are graphical tools used to analyze and understand data patterns easily.

Examples:

- Bar Chart: Compares categories
- Histogram: Shows data distribution
- Scatter Plot: Displays relationship between variables
- Box Plot: Identifies spread and outliers
- Pie Chart: Shows proportions

30. Mention any 3 types of data science models

Regression

Classification

Clustering

31. What is the confusion matrix?

A confusion matrix is a table used to evaluate the performance of a classification model by comparing actual and predicted values.

Components:

- TP (True Positive): Correctly predicted positive
- TN (True Negative): Correctly predicted negative
- FP (False Positive): Incorrectly predicted positive
- FN (False Negative): Incorrectly predicted negative

32. List out the software tools for EDA.

Python (Pandas, Matplotlib, Seaborn)

R

Excel

Tableau / Power BI

33. Differentiate between data science and data mining.

- Data Science:
A broad field involving data collection, analysis, modeling, and decision-making.
- Data Mining:
A subset of data science focused on extracting patterns from data.

Key Difference:

Data Science = complete process

Data Mining = pattern extraction only

34. List out the purpose of the project charter.

Gives formal approval to start the project

Defines project goals

Identifies project manager

Aligns team and stakeholders

35. Define Normalization.

Normalization

Real-time example:

In an e-commerce app, product prices range from ₹100 to ₹1,00,000.

To make ML models work efficiently:

₹100 → 0.0

₹1,00,000 → 1.0

This prevents large values from dominating smaller ones.

36.List out the Challenges in EDA

- 1.Handling Large Datasets
- 2.Interpreting Complex Patterns
- 3.Subjectivity

37.List out the need of Heatmap

- Helps to quickly identify:
 - Strong relationship
 - Weak relationship
- Useful for feature selection in data analysis

PART-B

1. Explain the steps involved in the Data Science Process with Real time examples.Give a Clear diagram for Data Science Process.
2. Briefly explain the architecture of data mining.
3. Briefly explain the architecture of data ware housing
4. How do you set the research goal, retrieving data and data preparation process in the data science process?
- 5.Explain the significance of Exploratory Data Analysis (EDA), how it helps in understanding and interpreting data, and describing the software tools and visualization techniques used in EDA.
- 6.Explain the fundamentals of Exploratory Data Analysis (EDA) with examples.

